SOEN 342: Software Requirements and Deployment

Iteration I: System Modelling and Requirements Specification

Fall 2026

Team members:

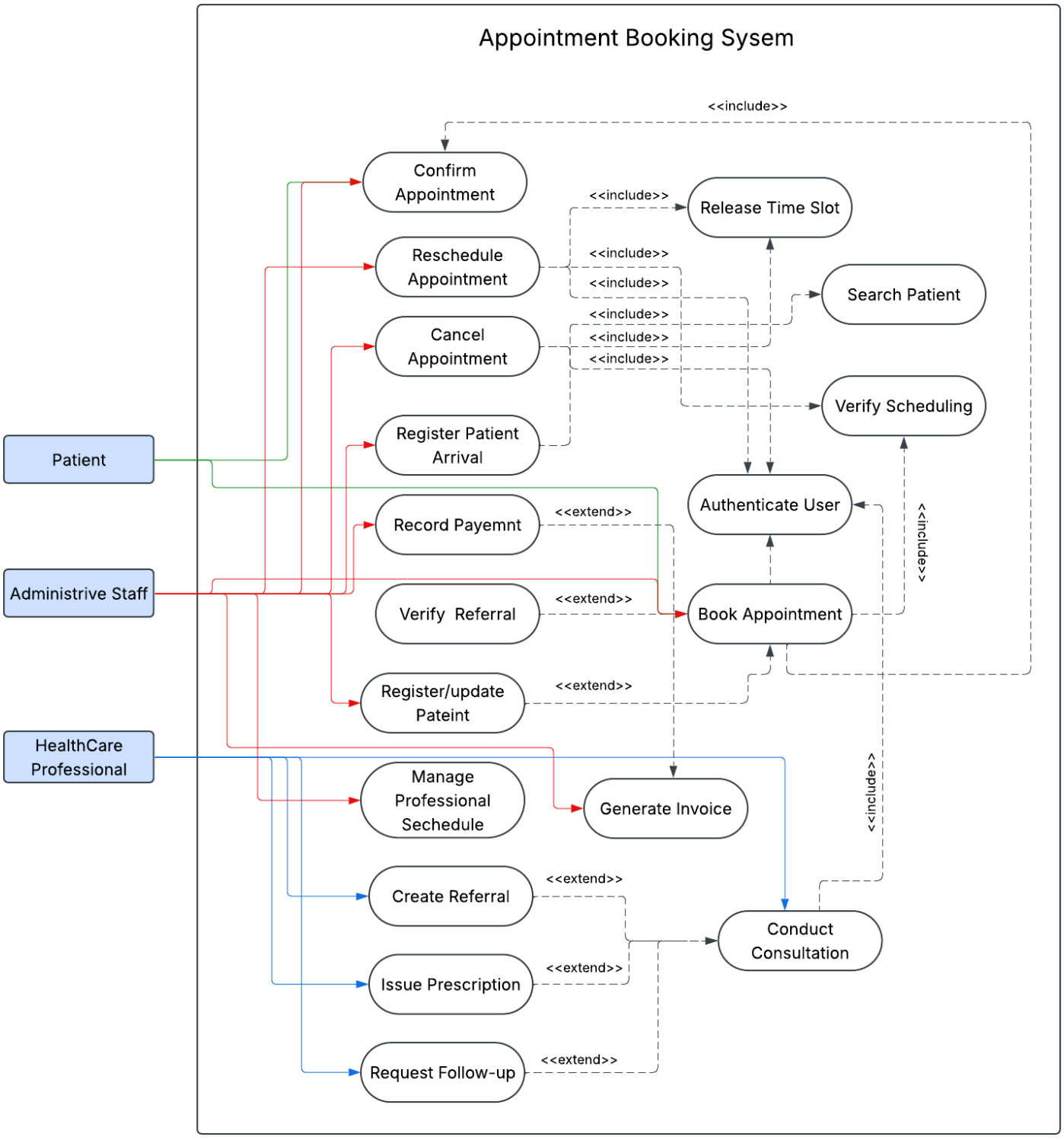
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Date: October 23, 2026

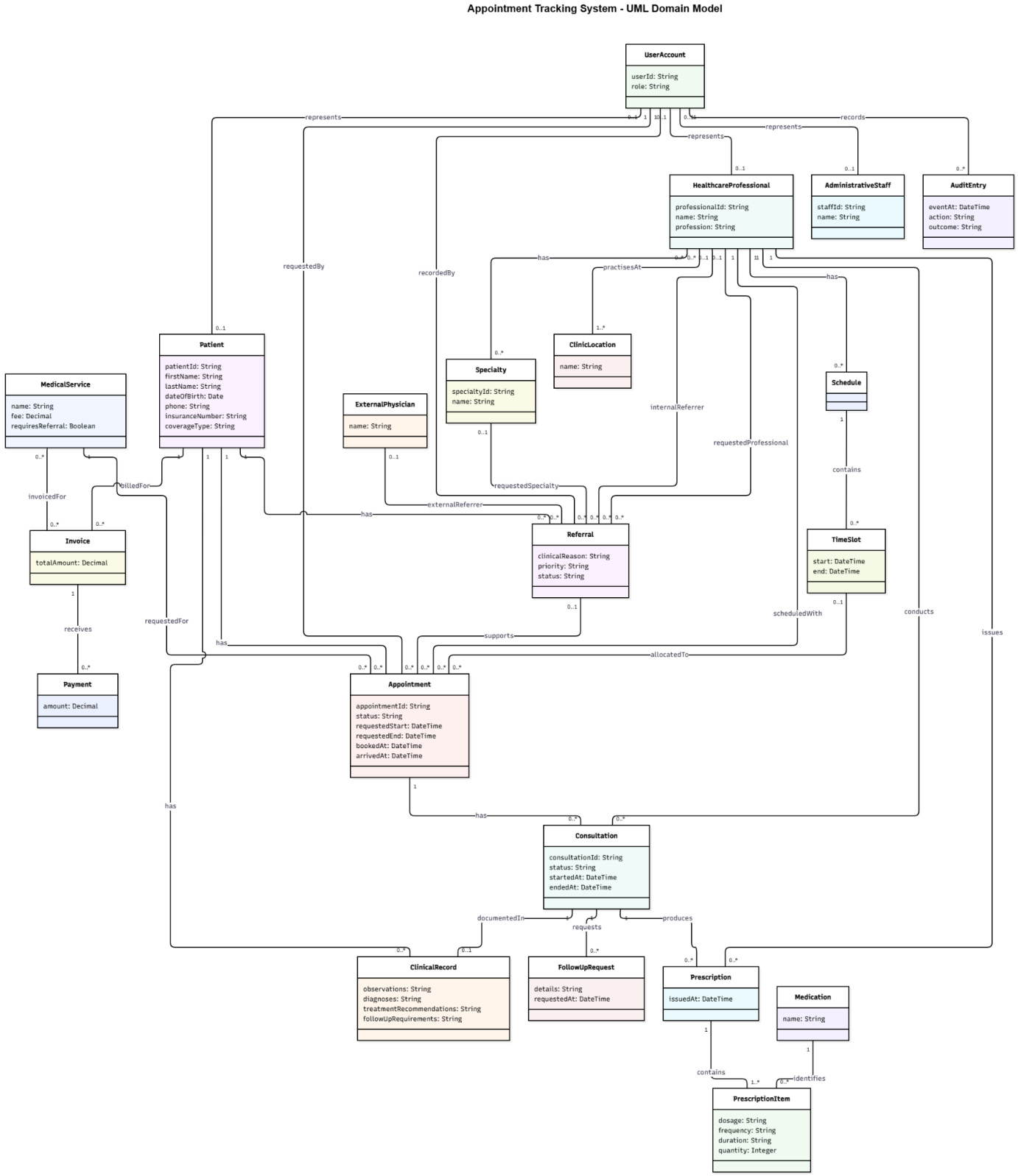
Use Case Diagram



use cases

ID	Use-case	Primary Actor	Critical - Type
UC-01	Book Appointment	Patients/Administrative staff	Critical
UC-02	Conduct Consultation	Healthcare Professional	Critical
UC-03	Create Referral	Healthcare Professional	Critical
UC-04	Search Patient	Administrative staff	non-critical
UC-05	Reschedule Appointment	Administrative staff	non-critical
UC-06	Cancel Appointment	Administrative staff	non-critical
UC-07	Register Patient Arrival	Administrative staff	non-critical
UC-08	Generate Invoice	Administrative staff	non-critical
UC-09	Record Payment	Administrative staff	non-critical
UC-10	Confirm Appointment	Administrative staff/Patient	non-critical
UC-11	Issue prescription	Healthcare Professional	non-critical
UC-12	Request a Follow up appointment	Healthcare professional	non-critical
UC-13	Register/Update Patient	Administrative staff	non-critical
UC-14	Manage Professional Schedule	Administrative staff	non-critical

UML Domain Model - Appointment Tracking System



Fully Dressed UC-01: Book Appointment

Field	Description
Use Case:	UC-01: Book Appointment (Critical operation)
Primary actor:	Patients requesting/booking appointments and administrative staff responsible for handling appointments
Stakeholders and interests:	- Healthcare professionals/Physicians: Wants accurate scheduling of appointments and fast referrals to other specialists and physicians. - Patients: Wants quick appointment books. - Administrative staff: Wants a simple/learnable UI and a fast/reliable system.
Preconditions:	The healthcare professional is available and the appointment meets scheduling constraints
Success Guarantee (Postconditions):	- Patient is associated to a scheduled appointment - If applicable the necessary referral is requested and later confirmed by the appropriate physician and/or healthcare professional.
Special Requirements:	- Web-accessible - Touch screen UI - Accessibility to all demographics (e.g. Elderly, visually impaired)
Technology and data variation list:	- Requests can be made via web browser or reception desk - Credentials can be shared through the patients RAMQ or private insurance
Open Issues:	- Do administrative staff have additional credibility when booking an appointment for a patient. (e.g. Overriding scheduling constraints)? - When an appointment is requested does it enter a queue to be reviewed or is it confirmed on the spot? - How are race conditions handled when two patients request an appointment at the same time slot?

Main Success Scenario: UC-01: Booking Appointment

Step	Who is performing the action	Description
1	**Patient or Clinic Staff**	Requests an appointment for a specific medical service, including the start and end times. Sets the appointments status to requested
2	**System**	Returns confirmation that the system received the request
3	**system**	Authenticates the user and determines their role
4	**System**	Returns confirmation of the users role and authentication
5	**System**	Verifies that the requested appointment doesn't conflict with specialists availability and scheduling constraints. Temporarily holds the time slot.
6	**System**	Returns confirmation that the requested appointment doesn't conflict with specialist(s) availabilities and scheduling constraints
7	**System**	Update the schedule to contain the newly booked appointment
8	**Patient or Clinic Staff**	Finalize the booking of the appointment. Set the appointments status to confirmed
9	**System**	Return a confirmation message and send an email containing the appointment details to the patient

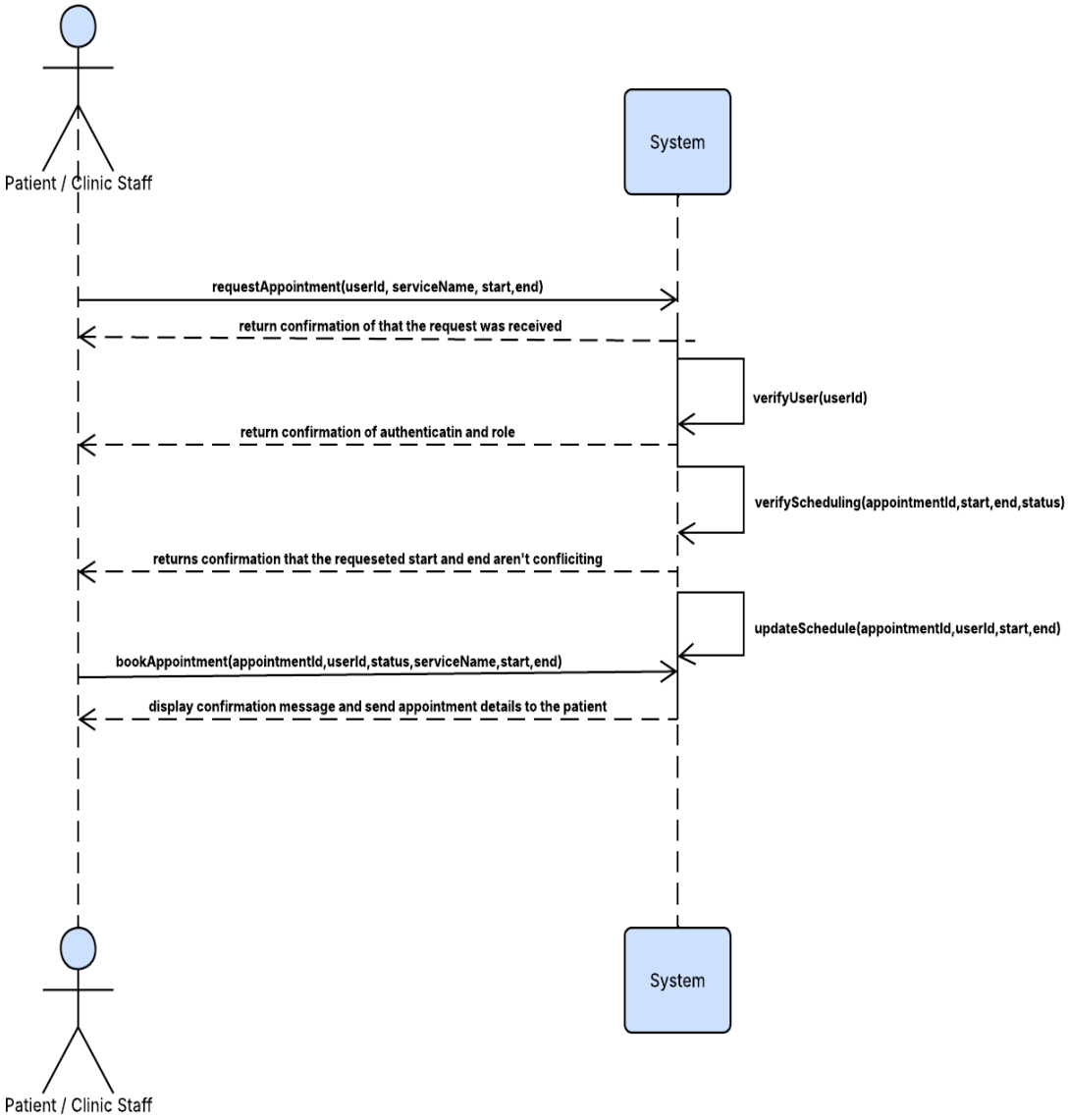
Extensions UC-01: Booking Appointment

Step	Condition	Outcome
1a	User enters faulty information (e.g. non-existent medical service, flipped start and end times)	System returns a message notifying the user of the error and prompts them to edit and resubmit their request.
3a	User's authentication and/or role isn't approved and they're denied access	System returns AccessDenied message, and denies the patient or clinic staff access to further pursue the appointment request.
5a	The appointment conflicts with the current schedule and doesn't meet the systems scheduling constraints	System returns FailedToSchedule error message and prompts the user to review his request information. Must re-submit their request.

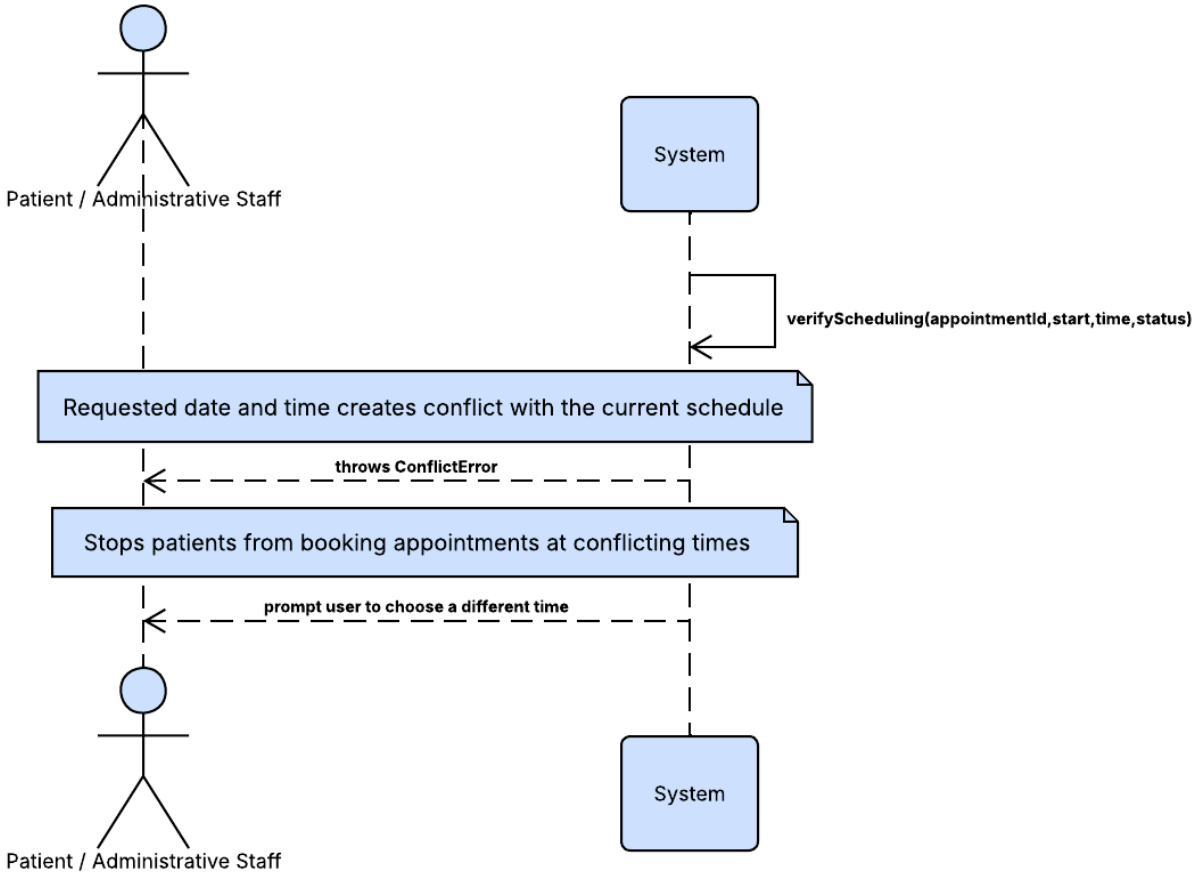
Failure Scenario: UC-01-F: Unavailable Healthcare Professional

Step	Who is performing the action	Description
1	System	Verifies that the requested appointment doesn't conflict with specialists availability and scheduling constraints.
2	System	Throws ConflictError as the request time conflicts with the healthcare professional's and or physician's schedule. Prevents appointments needing the same healthcare professional and/or physician from occurring at the same time.
3	System	Prompts user to choose a different date and time

UC-01: System Sequence Diagram - Main Success Scenario



UC-01: System Sequence Diagram - Failure Scenario



UC-01: System Operations

Contract ID	System Operation	Description
OC-01-01	requestAppointment(userId, serviceName, start,end)	Sends a request for an appointment to the system stating the medical service, start and end times. Sets appointment status to request. If the request was badly structured in any way (e.g. non-existent medical services, flipped start and end times) the system will return a message notifying the user of the error and prompt them to edit and resubmit their request.
OC-01-02	verifyUser(userId)	Authenticates the user and determines their role. Will deny further access if not approved.
OC-01-03	verifyScheduling(appointmentId,start,end,status)	Verifies that the requested start and end times don't conflict with availability and scheduling constraints. Temporarily holds the requested time slot if it passes. Otherwise, the user is prompted to choose a different start and end time.
OC-01-04	updateSchedule(appointmentId,userId,start,end)	Adds the appointment to the schedule according to the requested start and end time
OC-01-05	bookAppointment(appointmentId,userId,status,serviceName,start,end)	Finalizes the booking of the appointment. Changes the appointments status to confirmed.

UC-01: System Operation Contracts

OC-01-01: requestAppointment

Field	Description
Contract	OC-01-01
Operation	requestAppointment(userId, serviceName, start,end)
Cross-reference	UC-01 steps 1 and 2, extension 1a.
Preconditions	1. UserAccount, userId, exists 2. MedicalSerive, serviceName, exists
Postconditions	1. Appointment, appointmentId, is created (instance creation) 2. appointmentId is associated to patientId and staffId(association formation) 3. appointmentId.status is set to requested (attribute modification)
Failure / extension	The system will return a message notifying the user of the error and prompt them to edit and resubmit their request. Stops from misleading requests from being mixed into the set of scheduled assignments and/or slowing down/crashing the system.

OC-01-02: verifyUser

Field	Description
Contract	OC-01-02
Operation	verifyUser(userId)
Cross-reference	UC-01 steps 3 and 4, extension 3a.
Preconditions	1. requestAppointment was received without throwing any errors 2. UserAccount, userId, must exist 3. userId must have a valid role with access permissions
Postconditions	None
Failure / extension	If the user's Id or isn’t approved, the system will return an error message. Users will be denied any further access to the system.

OC-01-03: verifyScheduling

Field	Description
Contract	OC-01-03
Operation	verifyScheduling(appointmentId,start,end,status)

Cross-reference	UC-01 step 5, extension 5a.
Preconditions	1. Appointment, appointmentId, exists 2. appointmentId.status is set to requested 3. Start and end times don't conflict with current availability and scheduling constraints.
Postconditions	None
Failure / extension	System returns FailedToSchedule error message and prompts the user to review his request information. Must re-submit their request. Prevents appointments from occurring at the same time. Schedule constraints also maintain efficiency and reliability (no system crashes).

OC-01-04: updateSchedule

Field	Description
Contract	OC-01-04
Operation	updateSchedule(appointmentId,userId,start,end)
Cross-reference	UC-01 step 7.
Preconditions	1. verifyUser and verifyScheduling has already passed 3. AppointmentId, appointmentId, exists and is associated to userId 4. appointmentId.status is set to requested
Postconditions	1. A TimeSlot with start and end was created (instance creation). 2. The TimeSlot was added to the Schedule and associated with the appointmentId (association formation).
Failure / extension	If verifySchedule was not passed yet, the appointment can't be added to the schedule as it risks conflictions with availability and can contradict the current scheduling constraints.

OC-01-05: bookAppointment

Field	Description
Contract	OC-01-05
Operation	bookAppointment(appointmentId,userId,status,serviceName,start,end)
Cross-reference	UC-01 step 8.
Preconditions	1. appointmentId and userId exist and are associated to each other 2. appointmentId.status is set to requested 3. updateSchedule has completed

Postconditions	1. appointmentId.status is set to confirmed (attribute modification) 2. appointmentId.bookedAt is set to the current time (attribute modification)
Failure / extension	If the appointment doesn't have status, requestested, no further progress can be granted. A request for an appointment must first be made before it can be booked.

Fully Dressed UC-02: Conduct Consultation

Field	Description
Use Case:	UC-02: Conduct Consultation (Critical operation)
Primary actor:	Healthcare Professional responsible for the appointment
Stakeholders and interests:	- Healthcare Professional: documents consultation results efficiently and accurately - Patient: Confidential, accurate clinical records - Clinic: Accurate consultation history and proper logging of each clinical related read and write.
Preconditions:	The user is properly authenticated as a healthcare professional, an appointment exists for the patient.
Success Guarantee (Postconditions):	- The consultation's resulting clinical information are associated with the patient and their appointment. - If any prescriptions or follow up requests are made during the consultation, they are recorded.
Special Requirements:	- Only the assigned professional can read or write the appointments clinical information - All prescriptions are recorded internally and are not transmitted electronically to pharmacies.
Technology and data variation list:	None specified in system description
Open Issues:	None.

Main Success Scenario: UC-02: Conduct Consultation

Step	Who is performing the action	Description
1	**Healthcare Professional**	Requests the details of the appointment in order to document the consultation
2	**System**	Authenticates the healthcare professional to verify that they are able to access the appointment and obtain relevant patient information needed for the consultation.
3	**Healthcare Professional**	Starts documenting the consultation
4	**System**	Creates a new consultation record associated to the appointment
5	**Healthcare Professional**	Records all observations, diagnoses, treatment recommendations and follow-up requirements if needed/possible
6	**System**	Saves the clinical information in the consultation record
7	**Healthcare Professional**	Indicates that the consultation record is complete
8	**System**	Saves the completed consultation record and marks as completed.

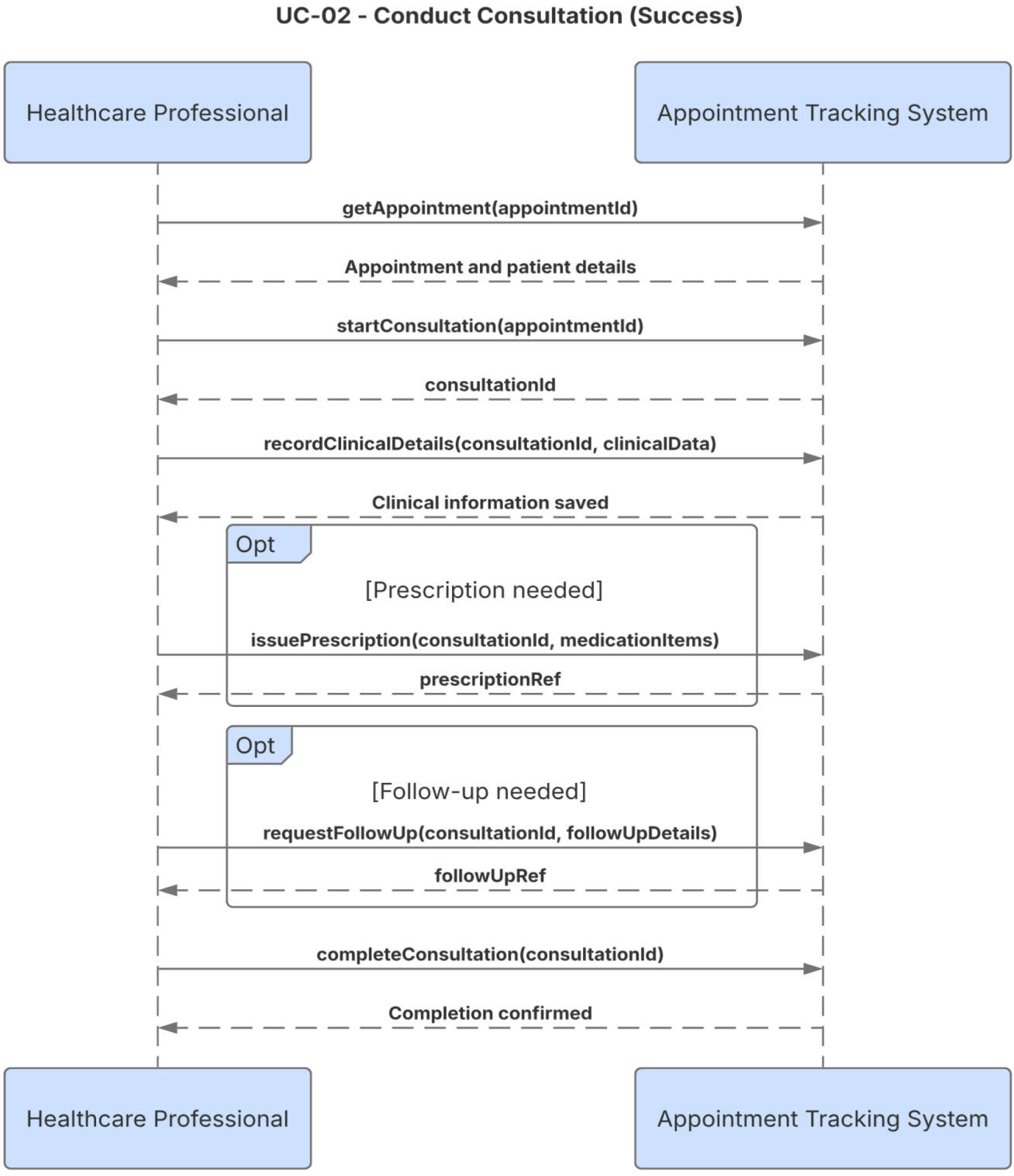
Extensions UC-02: Conduct Consultation

Step	Condition	Outcome
2a	The Healthcare professional is not assigned to the appointment, therefore not authorized.	System returns AccessDenied, and denies the user access to clinical information reading and writing. The denied access attempt is logged. Use case ends
2b	The appointment is not found or does not exist.	The system tells the user that the appointment cannot be found. Use case ends.
5a	The healthcare professional states that a prescription is needed	The system records the prescription, consultation continues.
5a1	The prescription information is invalid	The system rejects the prescription and prevents record creation. The healthcare professional can correct the mistake and continue or cancel
5b	The healthcare professional requests a follow up appointment	The professional provides the follow up request and the system records the request. Consultation continues
6a	The clinical information sent for submission is invalid or empty	The system reports the issue, refuses the save and allows the healthcare professional to correct and resubmit.

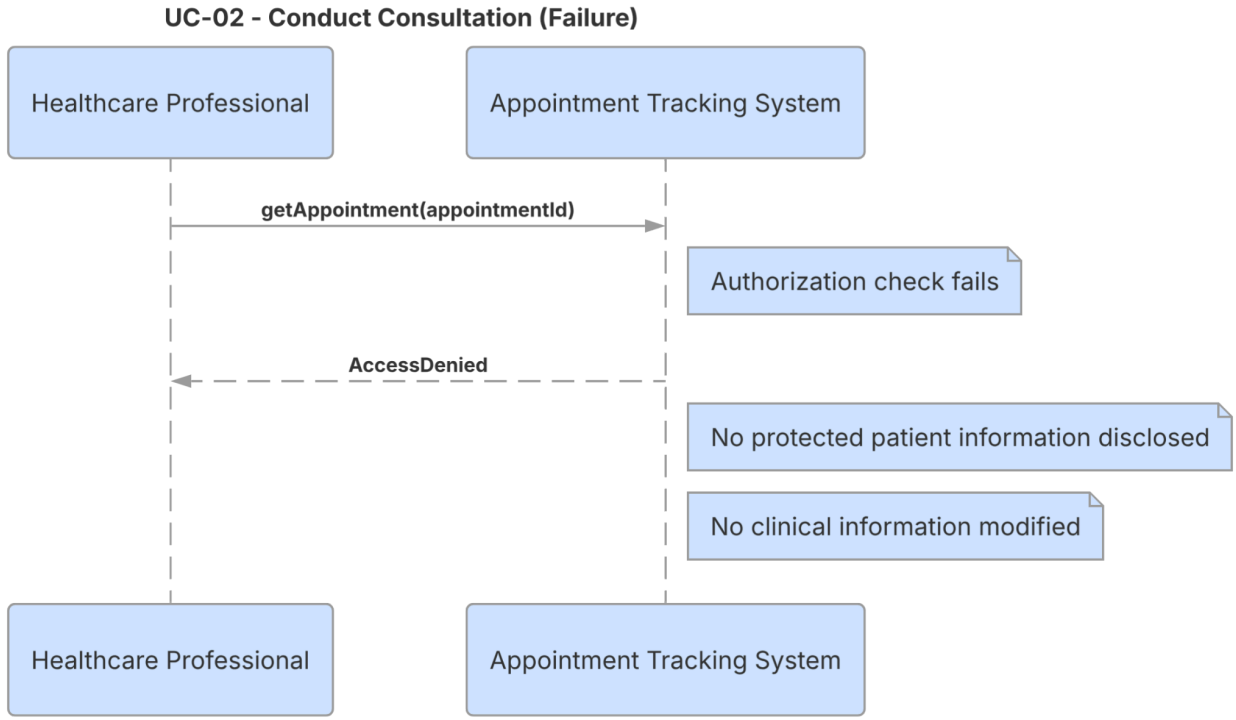
Failure Scenario: UC-02-F: Unauthorized Clinical Access

Step	Who is performing the action	Description
1	Healthcare Professional	Requests access to an appointment that they do not have access to.
2	System	Sees that the healthcare professional is not authorized and denies access. Ensures that it does not disclose any protected private information and records the denied attempt.

UC-02: System Sequence Diagram - Main Success Scenario



UC-02: System Sequence Diagram - Failure Scenario



UC-02: System Operations

Contract ID	System Operation	Description
OC-02-01	getAppointment(appointmentId)	access appointment and patient details after proper authentication is validated.
OC-02-02	startConsultation(appointmentId)	Create a consultation for the selected appointment ID.
OC-02-03	recordClinicalDetails(consultationId, clinicalData)	Save observations, diagnoses, recommendations and follow-up requirements.
OC-02-04	issuePrescription(consultationId, medicationItems)	Optionally record a prescription and medication items.
OC-02-05	requestFollowUp(consultationId, followUpDetails)	Optionally record a follow-up request.
OC-02-06	completeConsultation(consultationId)	Mark the consultation record complete.

UC-02: System Operation Contracts

OC-02-01: getAppointment

Field	Description
Contract	OC-02-01
Operation	getAppointment(appointmentId)
Cross-reference	UC-02 steps 1 and 2, extensions 2a and 2b, and UC-02-F.
Preconditions	1. An Appointment a identified by appointmentId exists. 2. The acting Healthcare Professional is authorized to access a
Postconditions	1. An AuditEntry e records the successful clinical read. [instance creation] 2. No Appointment, Consultation, or ClinicalRecord attributes are changed. [no attribute modification] 3. e is linked to the UserAccount being used. [association formation] 4. The permitted appointment and patient details are returned to the actor. [system response]
Failure / extension	If the appointment is not found then no details are returned. If authorization fails, the system returns AccessDenied, shares no protected private information, makes no clinical changes, and the denied attempt is logged in the system.

OC-02-02: startConsultation

Field	Description
Contract	OC-02-02
Operation	startConsultation(appointmentId)
Cross-reference	UC-02 steps 3 and 4.
Preconditions	1. Appointment a exists and the Healthcare Professional responsible is authorized to document the consultation.
Postconditions	1. A new Consultation c and an AuditEntry e are created. [instance creation] 2. c.status is set to InProgress and c.startedAt is recorded. [attribute modification] 3. c is linked to a and the responsible HealthcareProfessional. e is linked to the acting UserAccount. [association formation] 4. The consultationId is returned. [system response]

Failure / extension	If the appointment does not exist or the Healthcare Professional isnt properly authorized to handle the consultation, then no Consultation is created and no clinical information is modified.

OC-02-03: recordClinicalDetails

Field	Description
Contract	OC-02-03
Operation	recordClinicalDetails(consultationId, clinicalData)
Cross-reference	UC-02 steps 5 and 6 and extension 6a.
Preconditions	1. A Consultation c identified by consultationId exists and can be edited. 2. The actor is authorized to update c. 3. clinicalData contains valid, non-empty clinical information.
Postconditions	1. If no ClinicalRecord exists for c, a ClinicalRecord r is created and an AuditEntry e records the action. [instance creation] 2. The provided observations, diagnoses, treatmentRecommendations and or followUpRequirements are stored in r. [attribute modification] 3. r is linked to c if newly created and e is linked to the UserAccount being used. [association formation] 4. The system confirms that the information was saved. [system response]
Failure / extension	If submitted clinicalData is invalid or empty, the system must reject the save and allow correction and resubmission and no submitted clinical changes are committed.

OC-02-04: issuePrescription

Field	Description
Contract	OC-02-04
Operation	issuePrescription(consultationId, medicationItems)
Cross-reference	UC-02 extensions 5a and 5a1 and UC-11 Issue Prescription.
Preconditions	1. Consultation c exists and the actor is authorized to document. 2. The actor has prescribing privileges 3. medicationItems contains valid prescription information.

Postconditions	1. A Prescription p, its PrescriptionItem instances, and an AuditEntry e are created. [instance creation] 2. p.issuedAt and the dosage, frequency, duration and quantity of each item are stored. [attribute modification] 3. p is linked to c and the prescriber, and each item is linked to p and its Medication. e is linked to the UserAccount being used. [association formation] 4. A prescriptionRef is returned. The prescription is recorded internally and is not transmitted to a pharmacy. [system response]
Failure / extension	If prescription information is invalid, the system must reject it, create no Prescription or PrescriptionItem, and permit the professional to fix up the request or continue without prescribing.

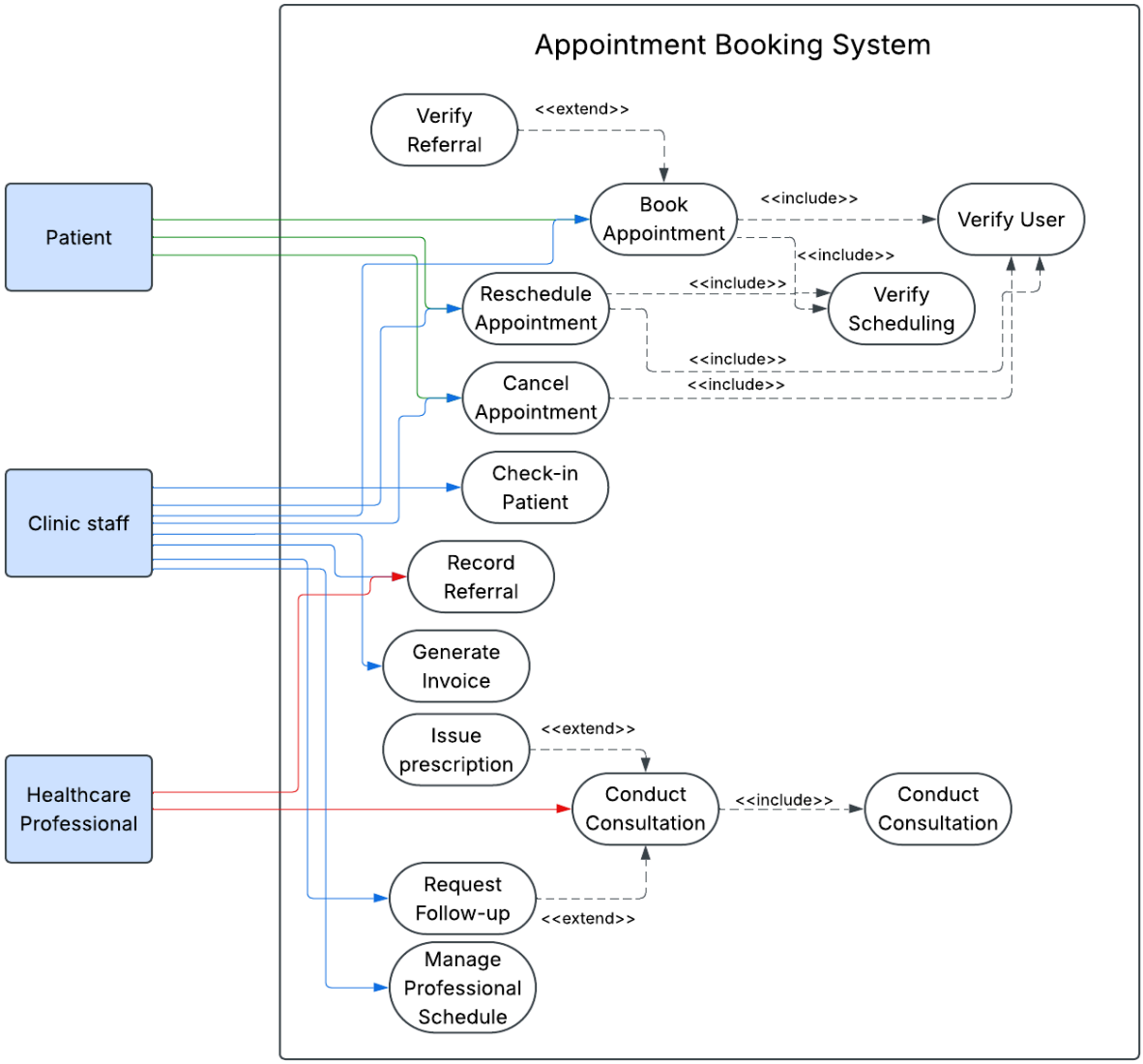
OC-02-05: requestFollowUp

Field	Description
Contract	OC-02-05
Operation	requestFollowUp(consultationId, followUpDetails)
Cross-reference	UC-02 extension 5b and UC-12 Request a Follow-up Appointment.
Preconditions	1. Consultation c exists and the actor is authorized to document. 2. followUpDetails provides a valid follow-up request.
Postconditions	1. A FollowUpRequest f and an AuditEntry e are created. [instance creation] 2. f.details and f.requestedAt are recorded. [attribute modification] 3. f is linked to c and e is linked to the UserAccount being used. [association formation] 4. A followUpRef is returned and this request does not itself book a new appointment. [system response]
Failure / extension	If the actor is not authorized or the request details are invalid, no FollowUpRequest is created and an error is then returned. The consultation is still available for continued documentation.

OC-02-06: completeConsultation

Field	Description
Contract	OC-02-06
Operation	completeConsultation(consultationId)
Cross-reference	UC-02 steps 7 and 8.

Preconditions	1. Consultation c exists and the actor is authorized to complete it. 2. c is open for documentation and its clinical information has been saved.
Postconditions	1. An AuditEntry e records completion. [instance creation] 2. c.status is set to Completed and c.endedAt is recorded. [attribute modification] 3. The Consultation remains linked to its Appointment, HealthcareProfessional and ClinicalRecord, and e is linked to the UserAccount being used. [association formation/retained associations] 4. The system confirms the completion of the operation. [system response]
Failure / extension	If c cannot be completed then it remains uncompleted and no completion changes are committed.


Appointment Booking System
Verify Referral
<<extend>>
Book Appointment
<<include>>
Verify User
<<include>>
Reschedule Appointment
<<include>>
Verify Scheduling
<<include>>
Cancel Appointment
<<include>>
Check-in Patient
Record Referral
Generate Invoice
Issue prescription
<<extend>>
Conduct Consultation
<<include>>
Conduct Consultation
Request Follow-up
<<extend>>
Manage Professional Schedule
Patient
Clinic staff
Healthcare Professional

use cases

ID	Use-case	Primary Actor	Critical - Type
UC-01	Book Appointment	Patients/Administrative staff	Critical
UC-02	Conduct Consultation	Healthcare Professional	Critical
UC-03	Create Referral	Healthcare Professional	Critical
UC-04	Search Patient	Administrative staff	non-critical
UC-05	Reschedule Appointment	Administrative staff	non-critical
UC-06	Cancel Appointment	Administrative staff	non-critical
UC-07	Register Patient Arrival	Administrative staff	non-critical
UC-08	Generate Invoice	Administrative staff	non-critical
UC-09	Record Payment	Administrative staff	non-critical
UC-10	Confirm Appointment	Administrative staff/Patient	non-critical
UC-11	Issue prescription	Healthcare Professional	non-critical
UC-12	Request a follow-up appointment	Healthcare professional	non-critical
UC-13	Register/Update Patient	Administrative staff	non-critical
UC-14	Manage Professional Schedule	Administrative staff	non-critical

Fully Dressed UC-03: Create Referral

Field	Description
Use Case:	UC-03: Create Referral (Critical operation)
Primary actor:	The referring healthcare professional and clinic staff
Stakeholders and interests:	- Referring professional: Wants to be able to accurately communicate the reason and priority of the referral. - Receiving professional: Wants sufficient information to process the referral. - Patients: Want an efficient and appropriate referral. - Administrative staff: Want a simple/learnable UI and referral information to process appointments.
Preconditions:	- The referring professional is authorized to create a referral. - The patient can be identified. - The required referral information is available.
Success Guarantee (Postconditions):	- A referral is created and associated with the patient, referring professional, and receiving professional. - The reason, status, and the priority of the referral are recorded.
Special Requirements:	- The clinical reason for the referral is visible only to authorized roles
Technology and data variation list:	- Patients can be searched for using multiple criteria
Open Issues:	- None

Main Success Scenario: UC-03: Create Referral

Step	Who is performing the action	Description
1	**Referring professional**	Searches for a patient for whom they wish to create a referral using insurance number, date of birth, phone number or name.
2	**System**	Displays the matching patients.
3	**Referring professional**	Selects a patient and initiates a new referral.
4	**System**	Verifies that the user is authorized to create a new referral and displays available receiving professionals.
5	**Referring professional**	Selects the required specialty or receiving professional and enters the reason and priority of the referral.
6	**System**	Validates the entered information and returns confirmation that the information is valid.
7	**System**	Creates and associates the referral with the patient and referring professional.
8	**System**	Records the requested professional, reason and priority.
9	**System**	Confirms the referral was created.

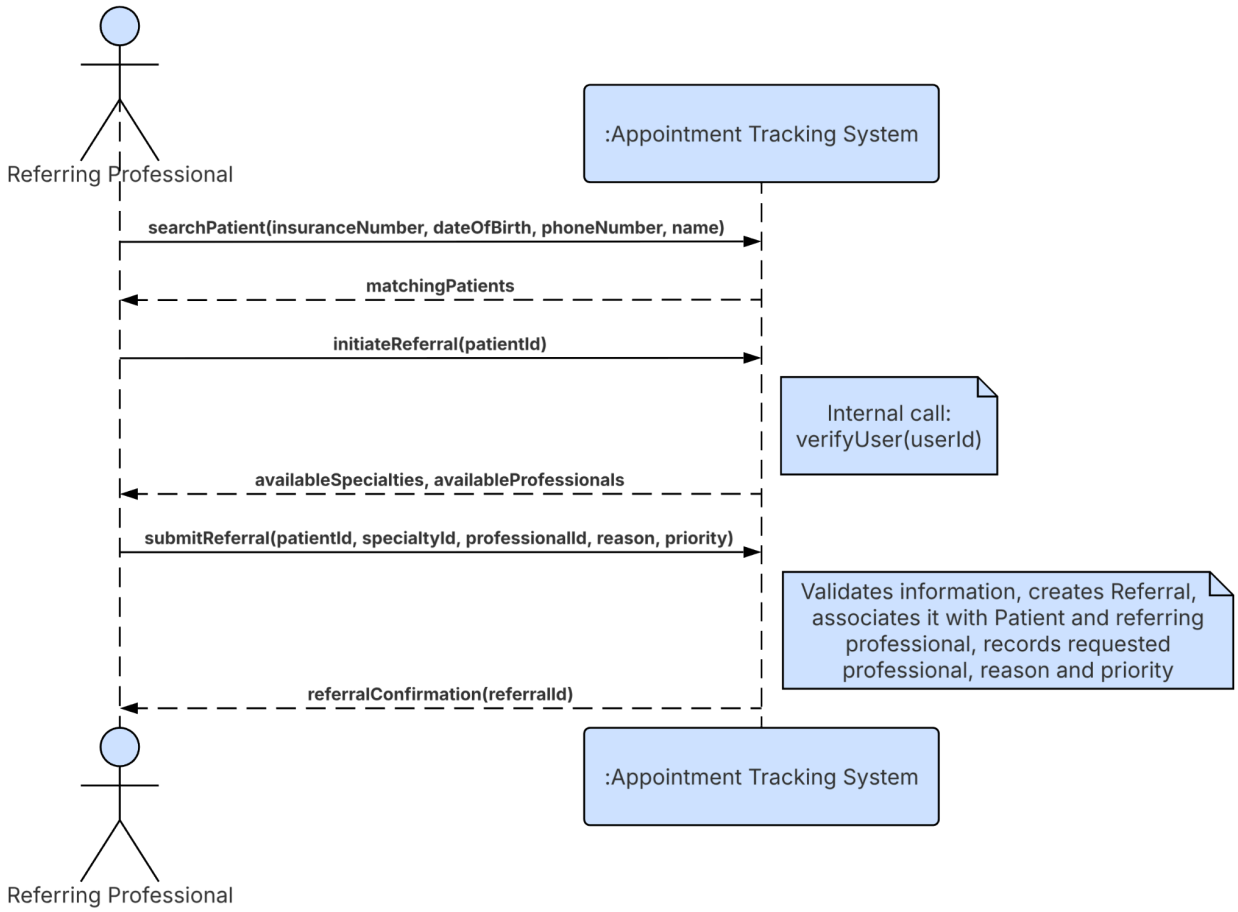
Extensions UC-03: Create Referral

Step	Condition	Outcome
2a	The requested patient for referral is not found	System returns a message notifying the user of the error and prompts them to search again or cancel.
3a	The referring professional is an external physician	Clinic staff must perform steps 1-9.
4a	The user is not authorized to create a referral	The request to create a referral is denied, and no clinical information is shared.
6a	The system determines that the required referral information is missing	System returns a message notifying the user that information is missing and prompts them to resubmit or cancel.
6b	The system determines that the selected specialty and receiving professional are invalid	System returns a message notifying the user that the selected specialty is invalid and prompts them to resubmit or cancel.

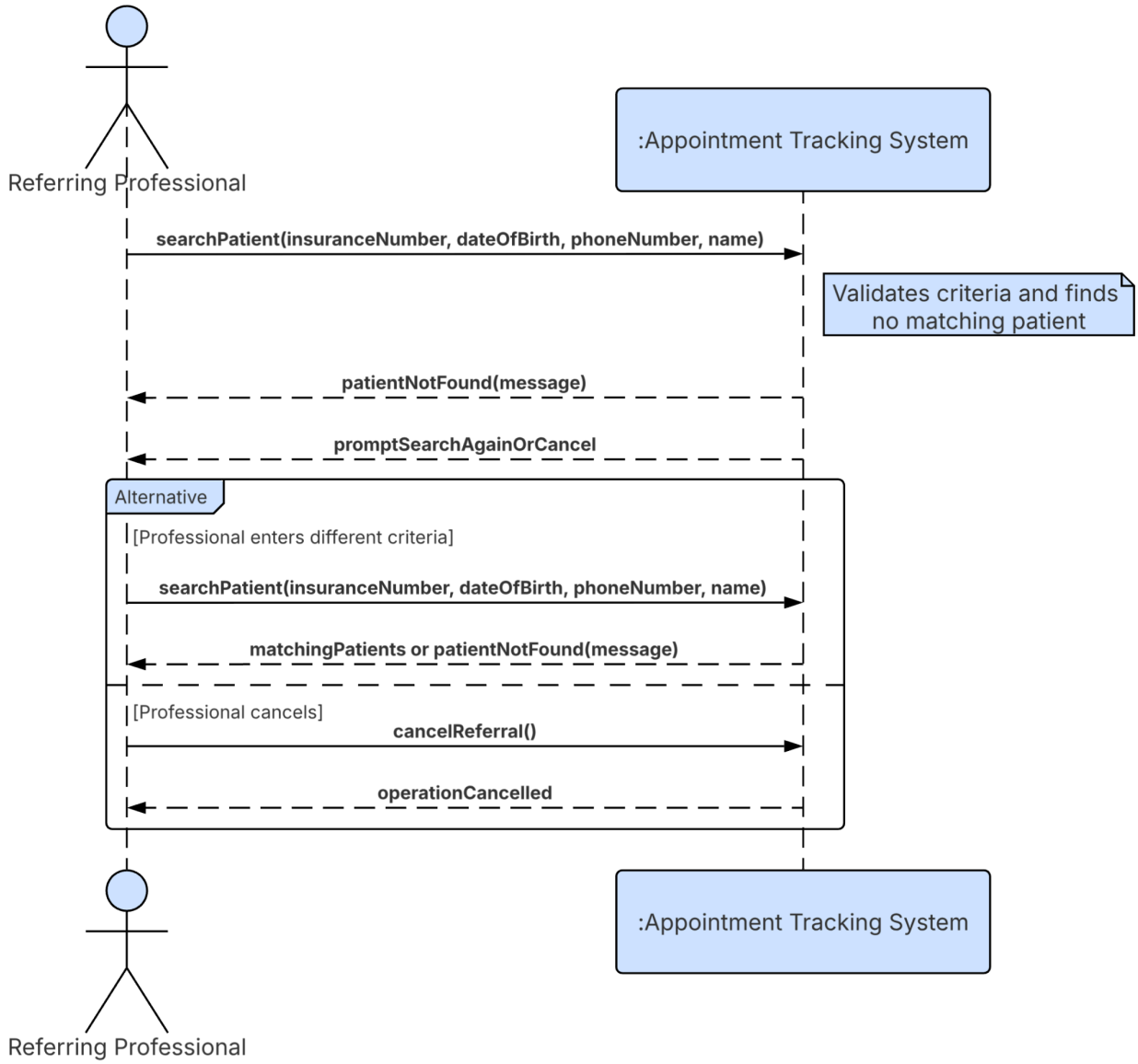
Failure Scenario: UC-03-F: Patient Not Found

Step	Who is performing the action	Description
1	**Referring professional**	Attempts to search for a patient to create a referral using insurance number, date of birth, phone number or name.
2	**System**	Validates the provided information and determines that no patients match the search requirements
3	**System**	Throws PatientNotFoundError as the requested patient's info cannot be found in the system. Returns a confirmation to the user that no patient was found.
4	**System**	Prompts the user to either input different search criteria or cancel the operation.

UC-03: System Sequence Diagram - Main Success Scenario



UC-03: System Sequence Diagram - Failure Scenario



UC-03: System Operations

Contract ID	System Operation	Description
OC-03-01	searchPatient(criteria)	Queries the system for any patients whose medical information contains any combination of the provided criteria (insurance number, date of birth, phone number or name) and returns the patient's ID. If the request was badly structured in any way (e.g. invalid date of birth) or no patient is found, the system will return a message notifying the user of the error and prompt them to edit and resubmit their request.
OC-03-02	verifyUser(userId)	Authenticates the user and determines their role. Will deny further access if not approved.
OC-03-03	initiate Referral(patientId)	Initiates the referral process and returns the associated patient's list of possible referral specialties and receiving professionals.
OC-03-04	createReferral(patientId, specialtyId, professionalId, reason, priority)	Creates the referral and associates it with the referring professional, receiving professional and patient. The system records the referral reason and priority and initializes the referral status to pending.
OC-03-05	cancelReferral()	Cancels the currently active referral. Deletes any created associations and recorded values. Returns the user to the start of the application

UC-03: System Operation Contracts

OC-03-01: searchPatient

Field	Description
Contract	OC-03-01
Operation	searchPatient(criteria)
Cross-reference	UC-03 steps 1 and 2, extension 2a.
Preconditions	1. Patient, patientId, exists 2. Patient has associated insurance number, date of birth, phone number and name.
Postconditions	1. patientId is returned to the user
Failure/extension	The requested patient is not found using the given criteria. The system will return a message notifying the user of the error and prompt them to edit and resubmit their request.

OC-03-02: verifyUser

Field	Description
Contract	OC-03-02
Operation	verifyUser(userId)
Cross-reference	UC-03 step 4, extension 4a.
Preconditions	1. UserAccount, userId, must exist 2. userId must have a valid role with access permissions
Postconditions	None
Failure/extension	If the user's ID isn’t approved, the system will return an error message. Users will be denied any further access to the system.

OC-03-03: initiateReferral

Field	Description
Contract	OC-03-03

Operation	initiateReferral(patientId)
Cross-reference	UC-03 step 3, extension 3a.
Preconditions	1. Patient, patientId, exists 2. Active user is authorized to create referrals
Postconditions	1. A list of available specialties and professionals is returned to the user
Failure/extension	If the referring professional is an external physician, these steps will be performed by clinic staff

OC-03-04: createReferral

Field	Description
Contract	OC-03-04
Operation	createReferral(patientId, specialtyId, professionalId, reason, priority)
Cross-reference	UC-03 step 5-8, extensions 6a and 6b.
Preconditions	1. patientId, specialtyId, professionalId, reason, and priority exist 2. Patient, ReferringProfessional and ReceivingProfessional exist 3. reason and priority are valid
Postconditions	1. An instance of Referral is created 2. Referral is associated with Patient, ReferringProfessional, and ReceivingProfessional 3. reason, priority and status are recorded in Referral
Failure/extension	The provided information for the referral is invalid or incomplete. The system throws an InvalidReferralError and notifies the user of the error. The system prompts the user to try again or cancel the operation.

OC-03-05: cancelReferral

Field	Description
Contract	OC-03-05
Operation	createReferral()
Cross-reference	UC-03 extensions 2a, 6a, and 6b
Preconditions	1. A referral has been initiated

Postconditions	1. All created associations and recorded values have been deleted 2. The user is returned to the start of the application
Failure/extension	None